

Question 1: What is Machine Learning?

Introduction:

Machine Learning is a part of artificial intelligence that lets computers learn from data without being programmed for every task. Algorithms improve automatically when they are exposed to more training data.

Types:

Machine learning has three types. Supervised learning uses labeled training data to make predictions. Unsupervised learning finds patterns in data without labels. Reinforcement learning uses rewards to train an agent.

Applications:

ML is used in image recognition, recommendation systems, and fraud detection. It is also used in natural language processing and speech recognition.

Conclusion:

Machine learning is an important field that helps computers learn from data. It has many applications in technology and is growing rapidly.

Question 2: Explain Neural Networks

Introduction:

Neural networks are computing systems inspired by the human brain. They have interconnected neurons in layers that process data and learn patterns.

Architecture:

A neural network has an input layer, hidden layers, and an output layer. Each neuron uses an activation function and has weights connected to other neurons. Deep neural networks have many hidden layers.

Training:

Neural networks are trained using backpropagation. The network makes predictions, calculates error using a loss function, and updates weights using gradient descent.

Conclusion:

Neural networks can learn complex patterns from data. They are used in deep learning for image recognition and language processing tasks.